



# JKBOSE 2026

## CLASS 11<sup>TH</sup>

### STRAIGHT LINES

MATHS

LECTURE-04

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# TOPICS TO BE COVERED

**1**

**Various Forms of the Equation of a Line**

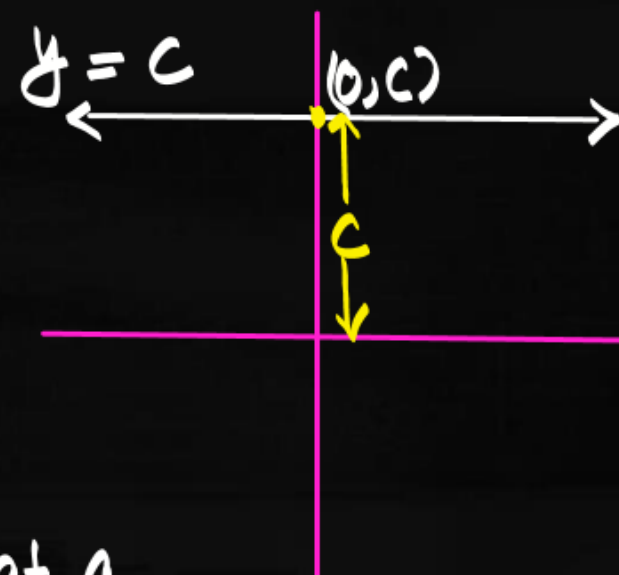
**2**

**Related Problems**

# Various Forms of the Equation of a Line

## \* Horizontal line:

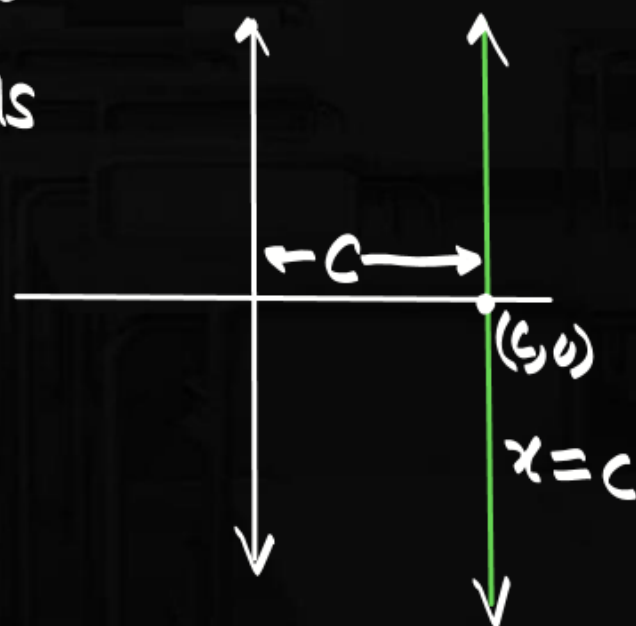
Line parallel to  $x$ -axis & at a distance of  $C$  units from  $x$ -axis



\*  $y = c$  (constant)

## \* Vertical line:

Line parallel to  $y$ -axis and at a distance of  $C$  units from  $y$ -axis



\*  $x = c$

## Point-slope form:

Suppose that the line  $l$  with slope  $m$  passes through the point  $A(x_1, y_1)$ . Then;

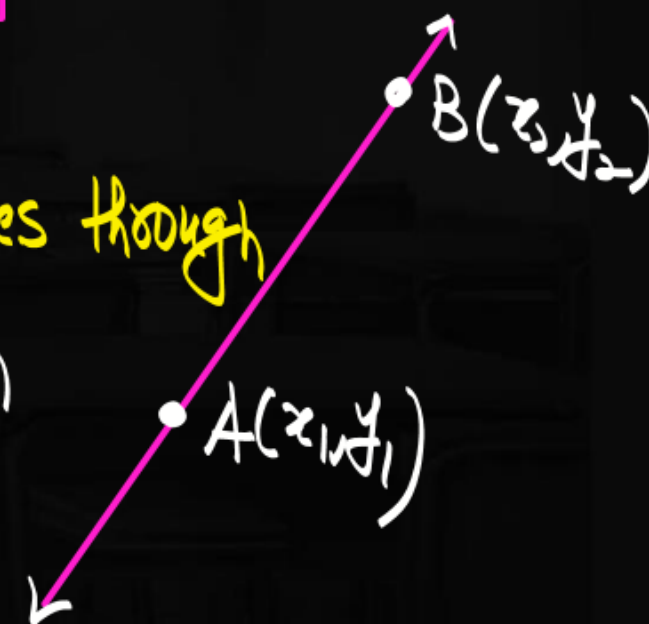


$$y - y_1 = m(x - x_1) \quad \text{--- (1)}$$

## \* Two-point form:

Suppose a line  $l$  passes through  $A(x_1, y_1)$  &  $B(x_2, y_2)$ . Then

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$



So from ①, we get

$$y - y_1 = \left( \frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1) \text{ (Two-point form)}$$

### \* Slope-Intercept form:

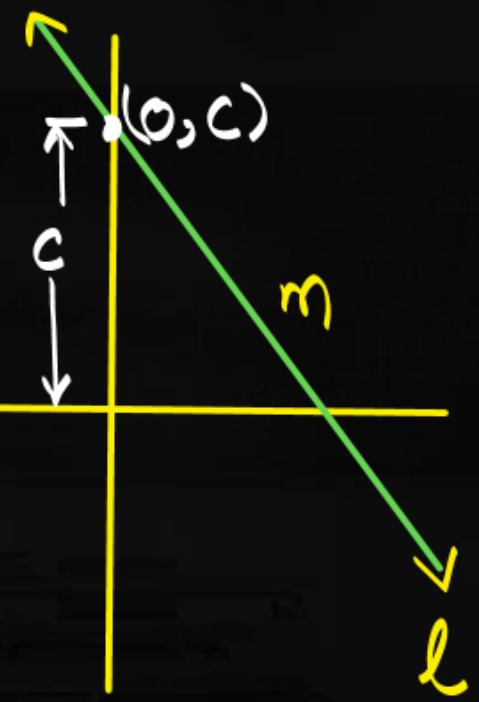
Suppose that a line with slope  $m$  has  $C$  as its  $y$ -intercept

So, from ①, we get

$$y - C = m(x - 0)$$

$$\Rightarrow y - C = mx$$

$$\Rightarrow \boxed{y = mx + C}$$



### Intercept form:

Suppose that a line  $l$  has  $a$  units as  $x$ -intercept &  $b$  units as  $y$ -intercept

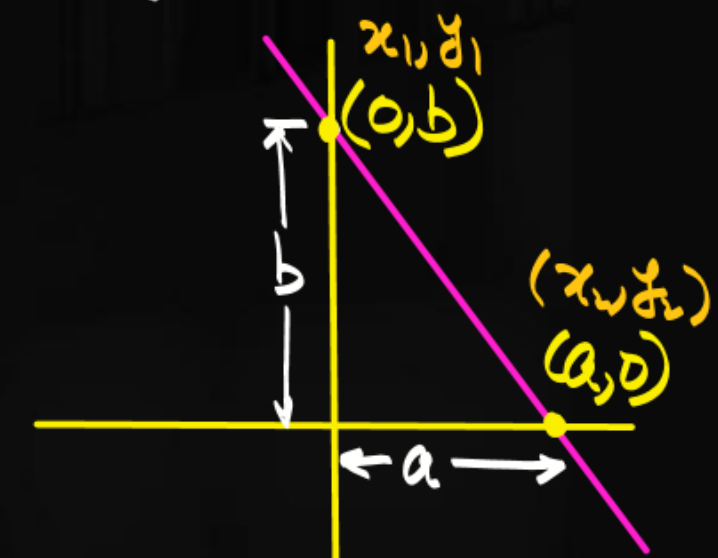
Then, the equation of  $l$  is:

$$y - b = \left( \frac{0 - b}{a - 0} \right) (x - 0)$$

$$\Rightarrow y - b = -\frac{b}{a}x$$

$$\Rightarrow \frac{y}{b} - \frac{b}{b} = -\frac{x}{a}$$

$$\Rightarrow \boxed{\frac{x}{a} + \frac{y}{b} = 1}$$



$$\begin{aligned} \frac{y}{b} - \frac{b}{b} &= -\frac{x}{a} \\ \frac{y}{b} - 1 &= -\frac{x}{a} \\ \frac{y}{b} + \frac{x}{a} &= 1 \end{aligned}$$

### General form:

The equation of the line in the general form is given by;

$$ax + by + c = 0 \quad \text{--- (1)}$$

$$\Rightarrow by = -ax - c$$

$$\Rightarrow y = \left(-\frac{a}{b}\right)x + \left(-\frac{c}{b}\right)$$

$$\text{So, } m = -\frac{a}{b}$$

$$\Rightarrow \text{slope of (1)} = \frac{-(\text{Co-efficient of } x)}{(\text{Co-efficient of } y)}$$

# EXAMPLE

$$y - y_1 = m(x - x_1)$$

Find the equation of the line through  $(-2, 3)$  with slope  $-4$ .

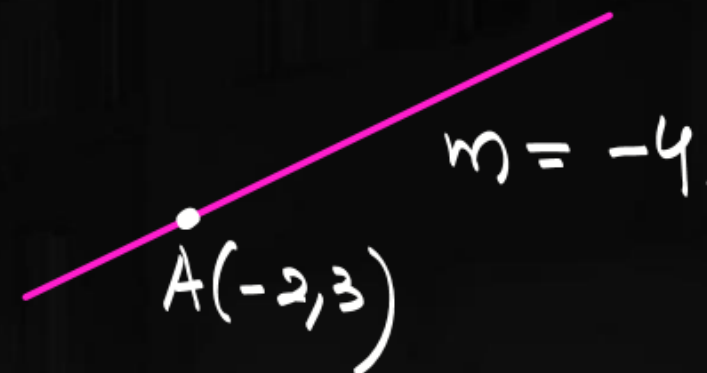
(Sol) The required equation is:

$$y - 3 = -4(x + 2)$$

$$\Rightarrow y - 3 = -4x - 8$$

$$\Rightarrow 4x + y + 5 = 0$$

Hence the result.



## EXAMPLE

$$y - y_1 = m(x - x_1)$$

Find the equation of the **Perpendicular bisector** of the line **segment** joining the points **A(2,3)** and **B(6, -5)**.

(Sol) We know that the product of the slopes of perpendicular lines is  $-1$ , so

$$m \left( \frac{-5-3}{6-2} \right) = -1$$

$$\Rightarrow m \times \frac{-8}{4} = -1$$

$$\Rightarrow m \times -2 = -1$$

$$\Rightarrow m = \frac{1}{2}$$

Also by Mid-point formula,

$$M(x, y) = M\left(\frac{6+2}{2}, \frac{-5+3}{2}\right)$$

$$\Rightarrow M(x, y) = M(4, -1)$$

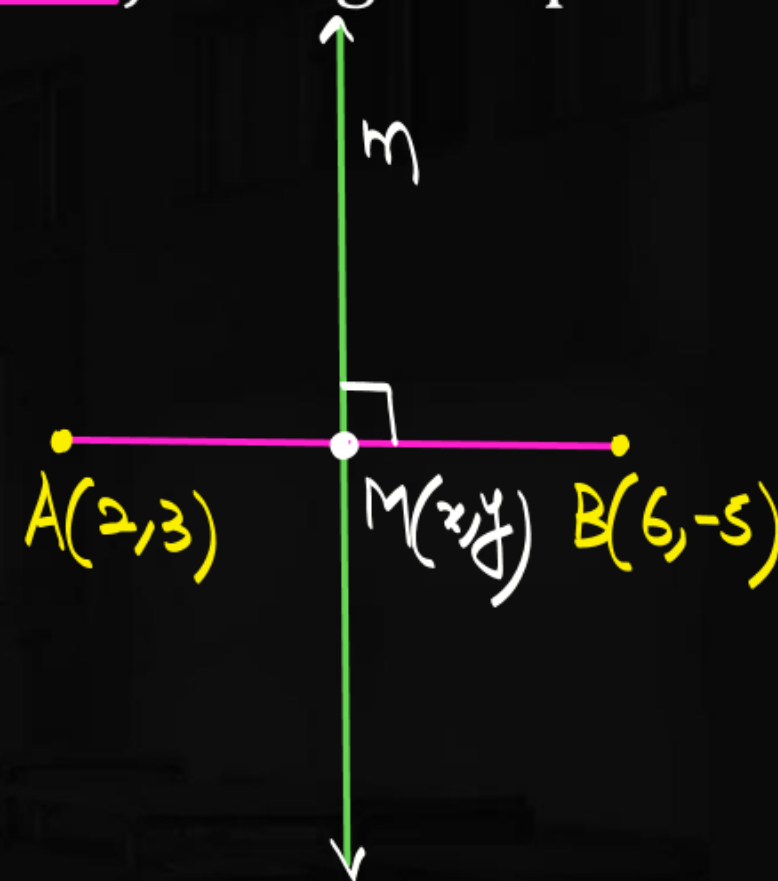
So, the required equation is;

$$y + 1 = \frac{1}{2}(x - 4)$$

$$\Rightarrow 2y + 2 = x - 4$$

$$\Rightarrow x - 2y - 6 = 0$$

Hence the result.



# QUESTION (NCERT EXAMPLAR):

A line cutting off intercept  $-3$  from the  $y$ -axis and the tangent <sup>of</sup> angle <sup>on</sup> of the  $x$ -axis is  $\frac{3}{5}$ , its equation is.

**1**  $5y - 3x + 15 = 0$  ✓

**2**  $3y - 5x + 15 = 0$  ✗

**3**  $5y - 3x - 15 = 0$  ✗

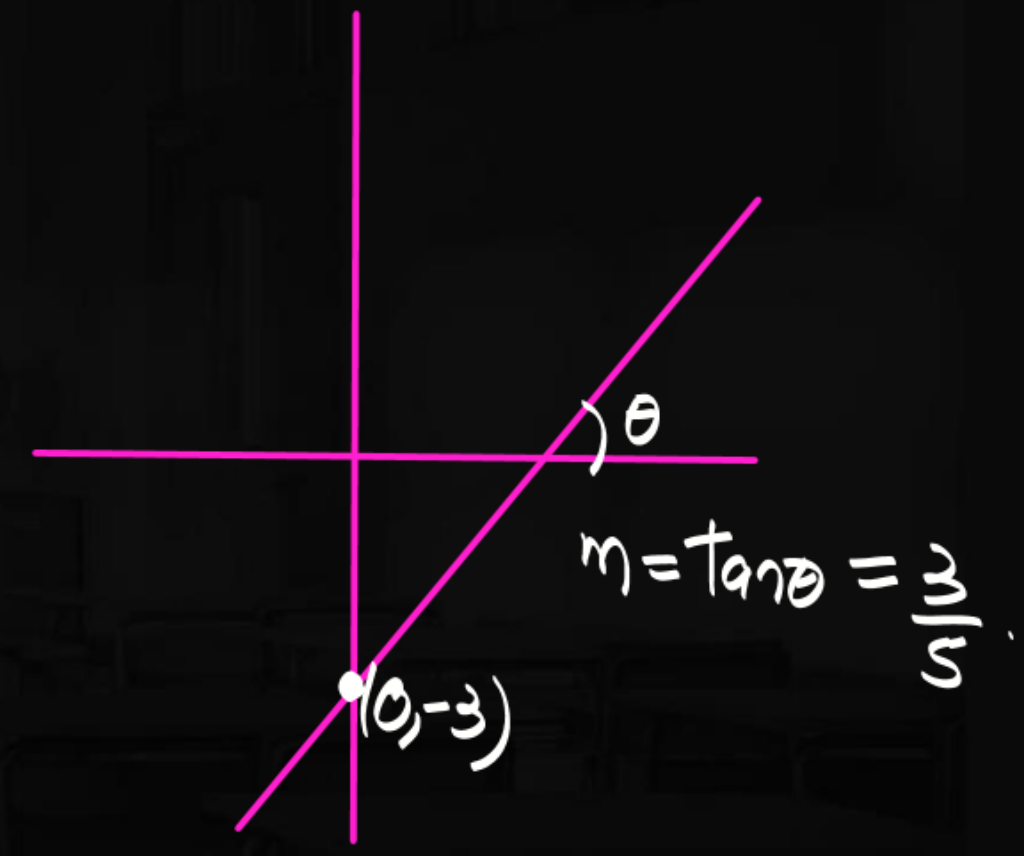
**4** None of these ✗

$$y - y_1 = m(x - x_1)$$

$$\Rightarrow y + 3 = \frac{3}{5}(x - 0)$$

$$\Rightarrow 5y + 15 = 3x$$

$$\Rightarrow 5y - 3x + 15 = 0$$



# QUESTION (NCERT EXAMPLAR):

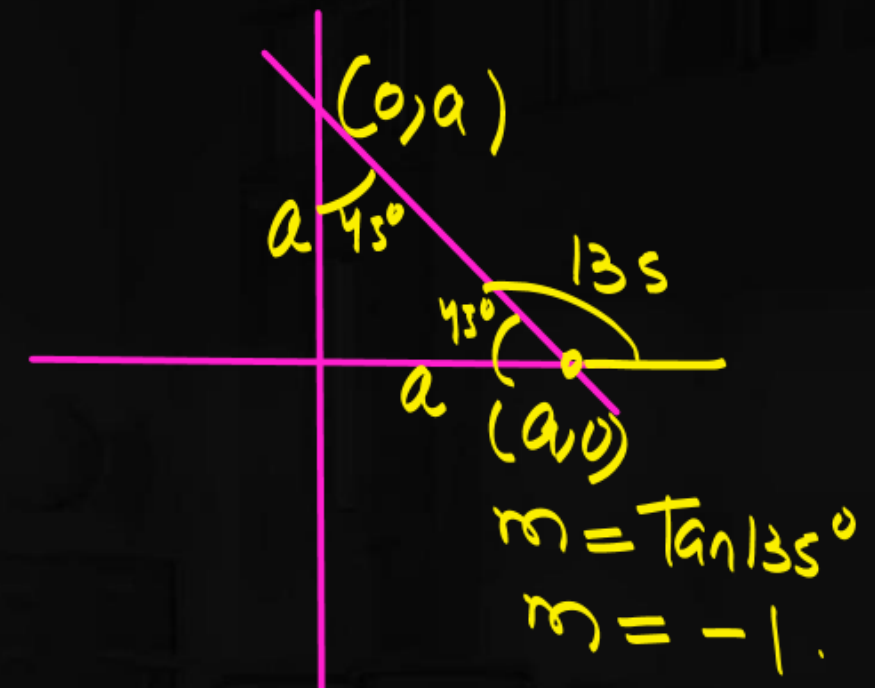
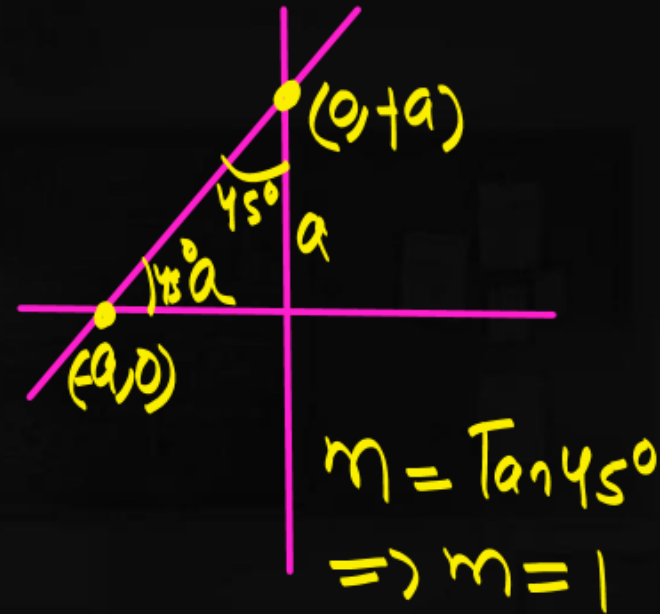
Slope of a line which cuts off intercepts of equal lengths on the axes is.

1  $-1$  ✓

2  $-0$  ✗

3  $2$  ✗

4  $\sqrt{3}$  ✗



**QUESTION (NCERT EXAMPLAR):**

The inclination of the line  $x - y + 3 = 0$  with the positive direction of  $x$ -axis is.

**1**  $135^\circ$

Homework  
problem.

**2**  $45^\circ$

**3**  $-45^\circ$

**4**  $-135^\circ$

**QUESTION (NCERT EXAMPLAR):**

For specifying a straight line, show many geometrical parameters should be known?

**1** 1 ✗

**2** 2 ✓

**3** 4 ✗

**4** 3 ✗

# QUESTION (JEE MAINS 2019):

If the straight line,  $2x - 3y + 17 = 0$  is perpendicular to the line passing through the points  $(7, 17)$  and  $(15, \beta)$ , then  $\beta$  equals:

**1**  $-\frac{35}{3}$  ✗

**2**  $5$  ✓

**3**  $\frac{35}{3}$  ✗

**4**  $-5$  ✗

(Sol) we know that

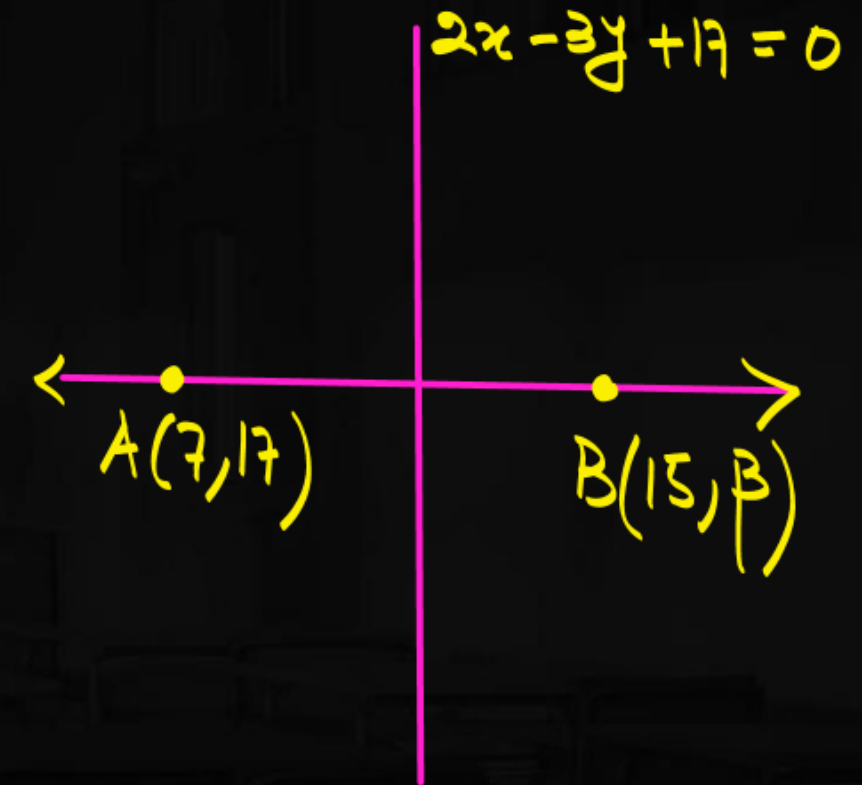
$$\left(\frac{-2}{-3}\right) \times \left(\frac{\beta - 17}{15 - 7}\right) = -1$$

$$\Rightarrow \frac{2(\beta - 17)}{3 \times 8} = -1$$

$$\Rightarrow \beta - 17 = -12$$

$$\Rightarrow \beta = -12 + 17$$

$$\Rightarrow \boxed{\beta = 5}$$



## QUESTION (NCERT EXAMPLAR):

The tangent of angle between the lines whose intercepts on the axes are  $a, -b$  and  $b, -a$ , respectively, is

Homework  
problem.

**1**  $\left| \frac{a^2 - b^2}{2ab} \right|$

**2**  $\left| \frac{b^2 - a^2}{2} \right|$

**3**  $\frac{b^2 - a}{2ab}$

**4** None of these

**Thank**  
**You**